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On acquiring data for large-scale crack bridging at high strain rates

C Lundsgaard-Larsen[1], R Massabò[2] and BN Cox[3]

Abstract

Numerical simulations are used to design test geometries and loading histories that are suitable for probing the large-scale bridging effects of through-thickness reinforcement that is shearing at high strain rates. The bridging effects are represented by a cohesive law and tests are sought that will determine any rate dependence in its parameters. The end-notched flexure test is studied, because it allows easy application of time-dependent loading and has proven to be an information-rich test in the quasi-static case. However, dynamic conditions greatly complicate fracture behavior, with possible regimes of hammering and multiple cracking, which should be avoided when maximum information is sought. Information content is addressed by focusing on regimes within the full computed solution space where crack growth is approximately steady state and the information content of experiments can be most easily assessed. Numerical results show that the hypothetical rate dependence in the cohesive law causes strong and measurable changes in the regime of steady-state behavior, if the tests are properly selected to vary the crack sliding speed. The estimates of information content are conservative, because the information available from all possible tests of specimens designed by analysis of the steady-state regime will necessarily exceed the information deduced by analyzing the steady-state regime alone.

Keywords

Dynamic delamination fracture, end-notched flexure test, dynamic cohesive law, computational modeling, through-thickness reinforcement, crack bridging, parameter identification

Introduction

Through-thickness reinforcement such as stitches or pins can increase the delamination resistance of laminated polymer or ceramic matrix composites by an order of magnitude or more.[1–9] The load required to propagate a delamination crack across fracture specimens can be raised by through-thickness reinforcement, which extends the precritical domain that precedes unstable crack growth, thus improving damage tolerance and ductility, that is, the capacity to undergo large deformations and displacements before failure.[10] These trends offer the materials designer significant opportunities for matching the performance of a material to specific applications, especially where damage tolerance and energy absorption are strongly desired. However, potential benefits can be preempted by adverse effects of the reinforcement on other failure mechanisms.[5] Therefore, careful selection of the parameters of the through-thickness reinforcement is needed, which must be based on experimental characterization of the performance of through-thickness reinforcement in relevant dynamic regimes. The nontrivial question of how to acquire such data is the subject of this study.

An expanding body of test data has now been published on the crack bridging effects of stitches and pins under static loading, both for single stitches or pins in model specimens and for arrays of stitches or pins bridging cracks in delamination specimens.[5,9–21] The observed mechanisms of deformation of the reinforcement under mixed-mode loading have been modeled with reasonable success, so that a good understanding exists of the main material factors influencing the efficacy of pins or stitches as bridging entities.[13,22,23] The bridging effect can be represented accurately in a

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nonlinear fracture model by cohesive tractions acting along a process zone. Under quasi-static loading, a cohesive law relates the tractions to the crack displacement; for instance, for a single-mode case, $\sigma = \sigma(\delta)$ where σ is the traction and δ the crack displacement. Accurate accounts of static delamination test data can be obtained by inserting the traction law into fracture simulations.

In addition to its benefits in static loading, through-thickness reinforcement has also been shown to successfully counteract the negative effects in dynamic delamination fracture and macrostructural behavior that arise from the inertia of the specimen and dynamic vibrations.[24,25] However, those and other theoretical studies assume that the constitutive behavior of the through-thickness reinforcement is approximately the same under dynamic conditions as it is under static conditions. This may not be the case. In fact, some indications already exist that show that rate effects should be expected in the bridging due to through-thickness reinforcement. Experiments in which fibers were pulled out of a substrate at rates of 1 and 100 mm/min show that the friction between the pulled out fibers and the laminate increases with speed.[26] The authors suggest that this is due to viscoelastic behavior of the fiber/matrix interface, coupled with a temperature increase due to friction. Theoretical studies of the micromechanics of fiber pullout also show that the inertia of the fibers will create rate effects in the fiber pullout problem.[27,28]

At low rates of loading and for slow crack growth, when inertia effects can be neglected, the problem of rate effects in the cohesive traction laws has been studied extensively in different material systems (see literature survey in Ref.).[29] At high strain rates and for fast crack propagation, the problem of rate effects in cohesive laws for conventional composite laminates has been studied only recently.[30] [Prior studies investigated rate effects on fracture toughness assuming a perfectly brittle material.][31–34] Furthermore, no attempt has been reported to determine the possible dynamic effects in the bridging tractions developed by through-thickness reinforcement by analyzing experiments. However, dynamic effects are expected when the bridging entities experience high strain rates (i.e., high rates of change of crack displacement).[27] This lack can frustrate optimal design because optimality does not necessarily result from inserting the maximum through-thickness reinforcement density. Although some properties increase monotonically as the density increases, cost, in-plane properties, and energy absorption can all be affected unfavorably.

The question, therefore, is what experiments will be most useful for acquiring data that determine the high strain rate behavior of through-thickness reinforcement. Rate effects enter the cohesive law through the local crack opening/sliding speeds along the process zone; for instance, for a single-mode case, $\sigma = \sigma(\delta, \dot{\delta})$, with $\dot{\delta} = \mathrm{d}\delta/\mathrm{d}t$ and t time. The ideal fracture tests would be tests where the fracture modes can be isolated (mode I or mode II) and the sliding/opening rates along the fracture surfaces kept constant in space and time and controlled via the applied load. Under such conditions, rate effects in the cohesive law could be defined through an inverse model using the results of a limited number of tests with different crack opening/sliding rates. For mode I, the double cantilever beam specimen loaded dynamically through the insertion of a wedge (flying wedge specimen) satisfies most of these requirements, as it ensures a regime of steady-state growth with a constant crack tip speed, equal to the loading speed, and time-invariant crack opening rates.[35] However, for mode II conditions, specimens satisfying the above requirements cannot be found; the fracture behavior is more complex.

This paper uses numerical studies to examine the suitability of dynamically loaded end-notched flexure (ENF) specimens to analyze rate effects in the cohesive mechanisms produced by through-thickness reinforcement. The ENF specimen has been shown to be an information-rich test for determining mode II cohesive laws in the quasi-static case.[10] In addition, when crack propagation in the ENF specimen is unstable in unreinforced laminates, even for quasi-static loading, the presence of through-thickness reinforcement promotes stable propagation and simplifies both testing and the solution of the inverse problem.[10] The test has a number of advantages over other mode II delamination specimens, such as the modified cracked lap shear test recently used by Guimard et al.,[30] to infer rate effects in unreinforced laminates: mode II conditions result from the symmetry of the problem and do not have to be forced using special experimental equipment, its geometry reflects that of actual structural components, and it allows an easier application of time-dependent loading. The desired information-rich ENF experiment should:

- generate dynamic crack propagation in the range of speeds where the effects of through-thickness reinforcement will be strongest, in particular, avoiding large crack speeds for which the process is dominated by the kinetic energy and the effects of the reinforcement become negligible[35].
- generate crack propagation in the domain where the process zone is developing and also where it is fully developed, thus maximizing the information content pertaining to all model parameters.
- avoid large flexural oscillations of the specimen as well as the development of high-order,

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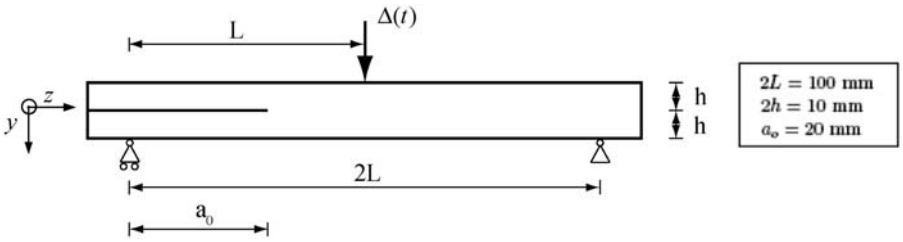


Figure 1. Dimensions of an ENF specimen and reference system.

high-frequency waves in the desired range of crack speeds.
- avoid the formation of multiple cracks in the specimen; matrix cracks and multiple delaminations may occur at the load point due to the rapid application of the load, in the initial stage of loading, or later at the unnotched end of the specimen.
- avoid failure by mechanisms other than delamination fracture.

Numerical studies are used in this paper to find such ENF experiments. Preliminary analyses performed with idealized rate-independent cohesive laws (Sections 'Results for quasi-static loading' and 'Results for high-speed loading') show that the dynamic crack propagation in the ENF specimen typically occurs under conditions that obscure the prospective information content of the experiments: crack tip speed typically changes continuously with time and crack sliding speed changes with time and along the process zone. These crack propagation characteristics will substantially complicate the solution of the inverse problem of deducing the rate-dependent cohesive traction laws from experimental data. However, the analyses also show the existence of crack propagation domains where crack growth is at least approximately steady state. In these domains, conclusions on rate effects can be inferred quite simply using numerical results obtained for rate-independent cohesive laws (Section 'Sensitivity to rate dependence in the cohesive law'). A simplified procedure is proposed that allows quick insight into the information content of candidate dynamic fracture experiments using the results of the simulations with rate-independent cohesive laws.

Characteristic length scales

The subject specimen is an ENF specimen, of length $2L$, thickness $2h$, and with a mid-plane notch of length a_0, loaded in three-point bending with $\Delta(t)$ the applied displacement, Figure 1. Delamination fracture from the initial notch is controlled by cohesive traction laws that approximate the fracture behavior of conventional and through-thickness reinforced laminates, as detailed in Section 'Choice of material properties, specimen geometry, and loading history'. Choosing an ENF configuration allows relation to the well-studied case of mode II fracture in the presence of large-scale bridging and to the length scales that control nonlinear fracture in delamination beams (slender bodies) under pure mode II loading.[10,30,36–38]

The small-scale bridging length scale, l_{SSB}, defines the length of the process zone of a crack propagating under small-scale bridging conditions, where the length of the process zone is constant and much smaller than the lengths of the crack and the ligament ahead of the crack tip, and the crack propagates with a constant fracture energy. For a globally homogeneous orthotropic body and rectangular cohesive law, with $\sigma = \sigma_0$ the cohesive tractions, δ_c the critical crack displacement beyond which the cohesive tractions vanish, and $G_b = \sigma_0\delta_c$ the cohesive fracture energy, the quasi-static small-scale bridging length is given by

$$l_{SSB} = \sqrt{a_{IIs}h}, \qquad a_{IIs} = \delta_c \frac{\bar{E}}{4\sigma_0} \tag{1}$$

where $\bar{E} = E/(1 - \nu_{zx}\nu_{xz})$ with E the longitudinal Young's modulus and ν_{ij} the Poisson coefficients. In thin bodies, the length of the process zone under mode II conditions scales with the square root of the thickness of the body h. In addition, a mode II delamination crack approaches the characteristic length from above as it propagates: the process zone length increases from zero at crack initiation to some value that is larger than the small-scale bridging limit and depends on the geometry and loading conditions, and then tends to the small-scale bridging limiting value. [The 'small-scale bridging condition,' as defined above, is referred to as the 'steady-state condition' in Ref. 36 to highlight differences between this limit in a slender beam, where l_{SSB} is a function of the thickness of the beam, and the small-scale bridging limit in an infinite medium, where l_{SSB} is a material property. The term 'steady-state fracture' will be used in this study with the meaning typical of dynamic fracture studies

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Table 1. Material properties for quasi-isotropic carbon/epoxy laminate related to coordinate system in Figure 1. The ultimate stresses σ_u in tension and compression are obtained from experiments presented in Ref. 42

E_z, E_x	50 GPa
E_y	10 GPa
$\nu_{zy}, \nu_{xy}, \nu_{zx}$	0.28
G_{zy}, G_{xy}	5 GPa
G_{zx}	$E_z/2(1+\nu_{zx})$
ρ (mass density)	1600 kg/m^3
$c_l = \sqrt{E_z/\rho}$	5590 m/s
σ_u (*tension*)	750 MPa
σ_u (compression)	630 MPa

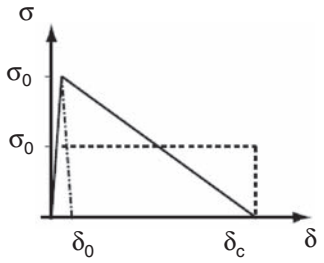


Figure 2. Trapezoidal and triangular cohesive laws used in the simulations.

Table 2. Parameters defining the three cohesive laws used in most simulations

	σ_0 (MPa)	δ_c (mm)	G_b (kJ/m^2)
Not reinforced (triangular)	20	0.05	0.5
Reinforced (triangular)	20	0.5	5.0
Reinforced (trapezoidal)	10	0.5	5.0

to refer to regimes of fracture, where the crack propagates with profile and speed that are time invariant. This limit configuration can be approached also under general large-scale bridging conditions (e.g., in the flying wedge specimen)].

Equation (1) accurately predicts the fracture zone length when the applied load can be represented by uniform shear tractions acting along the crack faces, the case for $a < L$ in an ENF specimen, where a is the crack length. For more complex loads, including the ENF specimen with $a \geq L$, variations can be expected in the process zone length. Variations in l_{SSB} of order unity will also arise when the shape of the cohesive law departs from the rectangular law used in deriving Equation (1).[39,40] Dynamic conditions introduce further variations in the zone length of Equation (1), which in steady-state propagation, decreases monotonically with crack tip speed.[30,41] However, if the crack tip speed is modest, for example, $v_{tip} < 0.3c_l$, where c_l is the longitudinal wave speed, variations are less than 5%.[30] Moderate velocities are of most interest for studying crack bridging because at higher speeds the kinetic energy of the fracturing material dominates crack propagation and bridging has no useful effect.[35] Thus, Equation (1) is only an approximate indicator of the zone length in small-scale bridging conditions, but is still useful in understanding trends in fracture.

Choice of material properties, specimen geometry, and loading history

The material is assumed to be a typical carbon/epoxy composite with quasi-isotropic lay-up and material properties as listed in Table 1.

Three idealized rate-independent cohesive laws are considered in the simulations, one representing a laminate without through-thickness reinforcement (triangular) and two representing laminates with through-thickness reinforcement (triangular and trapezoidal). Depending on the parameters of the through-thickness reinforcement, the mode II tractions may decline, remain approximately constant, or increase as the crack displacement increases from zero, until the reinforcement fails, after which the tractions must decrease (e.g., Ref. 13). Examining a decreasing triangular law and a rectangular law provides a simple study of the effects of such variations in the physics of fracture. The properties of the cohesive laws are illustrated in Figure 2 and listed in Table 2. The shapes of the laws and the parameter values in Table 2 are approximately consistent with those expected for laminates with z-pins or reinforcing rods or continuous stitches that fail and pull out of the laminate.[9,11,13] A fracture energy, given by the area below the traction-separation curve, $G_b = 5\,\text{kJ/m}^2$ and a critical displacement $\delta_c = 0.5\,\text{mm}$ have been used to describe laminates with through-thickness reinforcement, whereas $G_b = 0.5\,\text{kJ/m}^2$ and $\delta_c = 0.05\,\text{mm}$ have been used for laminates with no through-thickness reinforcement. The values of the maximum cohesive tractions, σ_0, in the two triangular laws for laminates with and without through-thickness reinforcement are made equal since they represent the critical strength of the unreinforced matrix; experiments show that introducing through-thickness reinforcement does not change this intrinsic matrix-dominated strength. The initial linear branch of the cohesive laws, which is necessary when using cohesive finite elements (Section 'Finite element model'), is chosen to be very stiff to limit its effect on the overall compliance of the specimen: $\delta_0 = 1.7$ and $3.3\,\mu\text{m}$ for the trapezoidal and triangular cohesive laws,

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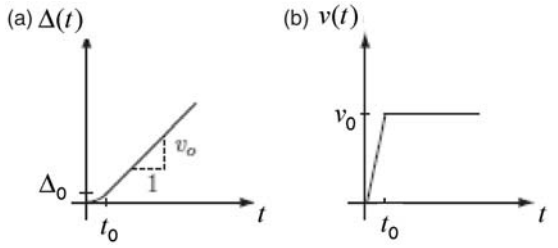


Figure 3. Schematic displacement (a) and velocity (b) histories for the load point in the simulations.

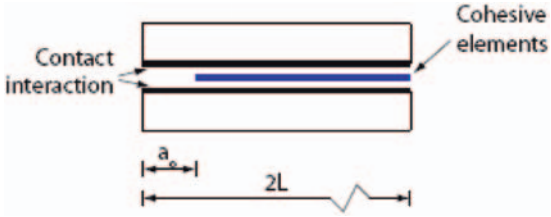


Figure 4. Exploded schematic illustration of an ENF specimen in the finite element model. Zero-thickness cohesive elements constitute the bonding between the beams, and contact interaction prevents interpenetration.

respectively. For the simulations performed in the last section of the paper and the derivation of behavioral maps, additional values have been used for the parameters σ_0, δ_0, and δ_c.

Specimen dimensions, $2L = 100\,\text{mm}$, $2h = 10\,\text{mm}$, and $a_0 = 0.4L = 20\,\text{mm}$ (Figure 1), were chosen in the simulations to take account of the anticipated length scales, to avoid failures by mechanisms other than delamination fracture at the mid-plane, namely compressive or tensile failure at the beam surfaces and multiple delamination, and for other considerations. First, the specimen should be longer than the expected process zone length to achieve steady-state propagation of a mature zone. However, if the specimen becomes very long this will lead to extremely large deflections. Simulations show that if the loading displacement is applied at a high acceleration at the midpoint of a very long slender specimen, the material close to the load point will move much faster than the material distant from the load point, which is constrained by inertia. This leads to a deflection shape remarkably different from the quasi-static case, and this nonequilibrium state will eventually lead to oscillatory behavior of the specimen. This effect, which is also present in an initial transient phase in short specimens, predominates when the specimen length is increased. For specimen thickness, a thicker arm limits the oscillatory behavior described above, reduces the maximum normal stresses at the upper and lower surfaces of the specimen, but increases the size of the process zone (Equation (1)). Finally, the precrack length should be short enough for the process zone to develop and reach a certain speed before the crack tip reaches the end of the specimen, yet it should be long enough to avoid unstable crack growth: reducing a_0/h increases the value of the critical load for the onset of crack propagation and, as a consequence, leads to more brittle postcritical responses (see, for instance, Ref. 10). Under quasi-static loading, the transitional value of the notch length for which the global response changes from stable to unstable is an inverse function of the length of the process zone in the small-scale bridging limit, l_{SSB}, and a direct function of the slenderness of the specimen, L/h.

In the simulations the specimen is loaded in displacement control. The loading history is an idealization of experimental loading conditions similar to those of a split Hopkinson pressure bar with a special pulse shaper.[43] The velocity of the load point is zero at time $t = 0$ and increases linearly to a constant speed v_0 over the time t_0 (Figure 3). If v_0 is small and t_0 large, the loading situation is quasi-static. If the velocity is large and t_0 approaches zero, the loading corresponds to a hard impact and the acceleration in the time interval 0–t_0 becomes large.

The ramp time t_0 has a large influence on the dynamic fracture behavior of the specimen. When the ramp time is short, high-frequency vibration modes may be excited, leading to an oscillating stress state near the crack tip. When the ramp time is large, crack propagation may occur before the applied loading speed has reached the final value and large values of crack tip speeds and sliding speeds may not be reached. The challenge is to choose a ramp time sufficiently small to obtain rapid crack growth, yet sufficiently large to avoid a highly fluctuating fracture behavior. Andrews et al.[24] analyzed the effect of ramp time on the dynamic amplification of the energy release rate for specimens with a stationary crack subject to a step force with a finite rise time. They found that if the ramp time was equal to the first natural vibration period T_1, there was no dynamic amplification and no subsequent oscillations. Conversely, smaller rise times led to oscillations in the specimen. By considering the fracture behavior of the tested specimens at various ramp times, the choice $T_1/2 \le t_0 \le T_1$ was found to be suitable compromises to obtain rapid crack growth, while limiting stress wave effects.

Finite element model

The 2D finite element model consists of two layers connected through zero thickness cohesive elements that partly cover the beam length (Figure 4). The precrack consists of the part of the beam without cohesive elements. Using a single cohesive interface constrains the

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cracks to form and propagate along the mid-plane. While this will always be the case when the specimen is loaded quasi-statically, also due to the presence of the notch, under dynamic loading conditions matrix cracks and multiple delaminations could form under the applied load due to its rapid application in the early stages. This damage mechanism, which is typically observed only for high impact energies,[44] is not included in the model; it could be studied using a model with multiple interface layers if experimental results warrant it.[45]

A frictionless contact interaction is formulated in a penalty master–slave configuration between the two layers to prevent interpenetration. The mesh in the layers is uniform, with square elements of size $0.1h$ with a refinement near the interface with a ratio of four between the side length of the smallest and largest elements. Nodes between bulk and cohesive elements are coincident and the side length of the cohesive elements is $0.025h$; this length satisfies the requirement that the cohesive zone should span several cohesive elements at all times during the simulations. Mesh independence was verified with a convergence study.

In a cohesive interface model, there is no crack tip singularity. The position of the crack tip, or leading edge of the process zone, is identified as the point where damage starts, namely the point where the cohesive element displacement equals δ_0 and the cohesive traction–displacement relation deviates from the initial linear regime. The trailing edge of the process zone is the point where the cohesive element displacement equals δ_c (Figure 2).

The model was implemented in the commercial finite element code ABAQUS 6.6, which solves the problem in the explicit domain and includes inertia effects. The four-node plane strain element CPE4R and four-node element COH2D4 are used for the layers and the interface, respectively. The density of the interface elements is chosen so that the total mass of the interface is 1/10 of the total mass, which is small enough not to add extra mass to the system and large enough to limit any effects on the automatically calculated time step. The implementation of the cohesive laws in ABAQUS is based on the damage model proposed by Camanho et al.,[46] which defines the mode I and mode II cohesive tractions as functions of a single damage variable (coupled model). As mode I displacements at the interface are always very small in all cases examined, the model is essentially equivalent in this problem to a classical uncoupled cohesive zone model.

The first natural vibration period, T_1, which is used to define the loading ramp, was defined through an eigenvalue analysis of the specimen where the cohesive elements are given a very large stiffness in the normal direction and a negligible stiffness for tangential deformation to allow sliding but restrain normal opening of the precrack (constrained vibration model). For specimens of length $2L = 100$ and 210 mm, the oscillation periods are $T_1 = 188$ and 588 μs, respectively.

Results for quasi-static loading

It is beneficial first to consider the results from quasi-static loading to distinguish between static and dynamic effects. In the quasi-static analyses, in this study, a loading rate $v_0 = \mathrm{d}\Delta/\mathrm{d}t = 0.05\,\mathrm{m/s}$ was found to be sufficiently slow and a convergence analysis showed no visible dynamic effects.

Laminate without through-thickness reinforcement

In Figure 5, the reaction force vs. applied displacement, and the positions of the leading and trailing edge of the crack vs. applied displacement, are plotted for quasi-static loading of a specimen with no through-thickness reinforcement ($G_b = 0.5\,\mathrm{kJ/m^2}$ and $t_0 = T_1$). Initially, the force increases linearly up to a load point displacement of 0.2 mm, at which point the leading edge starts to propagate. Propagation is stable up to an applied displacement of 0.55 mm. Thereafter, the leading edge of the crack propagates unstably up to the center of the specimen, which entails a snap-back instability described by a sudden drop in the applied load and small oscillations in the reaction force history; simultaneously, the trailing edge of the crack starts to propagate. Propagation of the leading crack tip into the second half of the specimen occurs by progressively increasing both the applied displacement and the reaction force. [Quasi-static ENF tests on unreinforced laminates are typically arrested when the crack, after a phase of unstable propagation, approaches the center of the specimen. In this study, numerical results are presented also for longer cracks as they highlight interesting features of the response.] The dashed line in Figure 5(a) represents the calculated load–displacement relation of the fully cracked specimen. This is calculated from shear deformable beam theory by assuming two center-loaded, simply supported beams each of thickness h. As expected, the load–displacement relation approaches this line as the crack length increases to span the whole length of the specimen.

The process zone length (found as the distance between the leading and trailing edges of the process zone, i.e., the vertical distance between curves in Figure 5(b)) is shown in Figure 6 as a function of load point displacement. A maximum process zone length of 13.5 mm is reached just before the unstable crack growth occurs. The zone length decreases monotonically throughout the remaining part of the simulation. The estimate of the small-scale bridging limit is given

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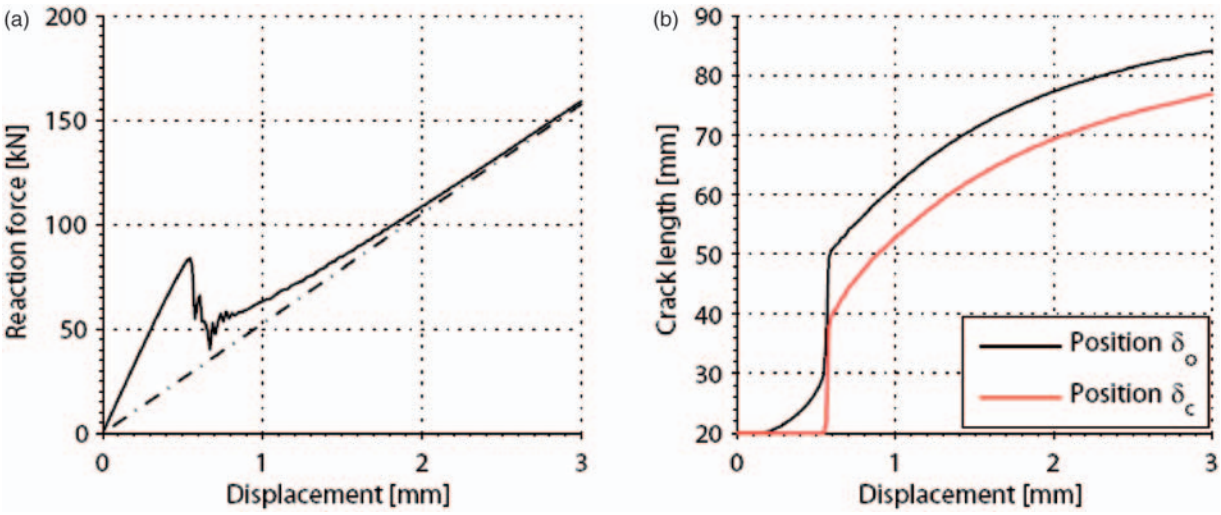


Figure 5. Results for quasi-statically loaded ENF specimen with $2L = 100$ mm and cohesive law corresponding to triangular with no reinforcements ($\delta_c = 0.05$ mm and $G_b = 0.5$ kJ/m^2). In both subplots, the horizontal axis represents the load point displacement (see Figure 1). (a) Reaction force at loading point and (b) position of leading (black line) and trailing (red line) edge of the process zone. The dashed line in (a) is the load–displacement curve of a completely damaged specimen (two beams each of thickness h) calculated from shear deformable beam theory.

by $l_{SSB} = \sqrt{\delta_c \bar{E} h/(4\sigma_0)} = 13$ mm, from Equation (1). The curve in Figure 6 tends to a smaller horizontal asymptote. This is due to the fact that a triangular cohesive law has been used and the concentrated reaction force at the midspan generates nonuniform stress and displacement fields at the bond line when $a \geq L$, whereas Equation (1) was derived for a rectangular law assuming uniform fields.

Under the load point and elsewhere along the crack when the crack tip has propagated beyond the center of the specimen, the crack surfaces are in compression. In unreinforced laminates, strong friction effects then arise. However, in the presence of through-thickness reinforcement, fracture surface contact is unlikely to occur: both experiments and analyses for stitches and pins, for example, show that, when a stitch or pin deforms in shear across the crack, its bulk tends to prop open the crack.[10] In fact, achieving purely mode II loading, with zero opening displacement, is extremely difficult experimentally. A specimen must be confined in a rigid test jig especially designed to prevent opening, and even then small opening displacements arise.[9,11] Therefore, a good simplifying assumption for the present analysis is that small opening displacements always exist and friction between the fracture surfaces can be neglected.

Laminate with through-thickness reinforcement

Results for the cases representing laminates with through-thickness reinforcement are shown in Figure 7. Dashed lines represent the trapezoidal cohesive law, whereas solid lines represent the triangular. By considering the reaction force in Figure 7(a), one sees that, in contrast to the case without through-thickness reinforcement, unstable crack growth does not occur, due to the stabilizing effects of a large process zone. This behavior confirms experimental observations on through-thickness reinforced laminates in Ref. 10. Consider the crack length diagram in Figure 7(b): in the specimen with a triangular cohesive law (solid line), the crack tip (leading edge) extends from the precrack at a loading displacement of 0.2 mm and the process zone subsequently builds up. As the crack tip passes the center of the specimen (i.e., the load point corresponding to 50 mm on the vertical axis), the curve

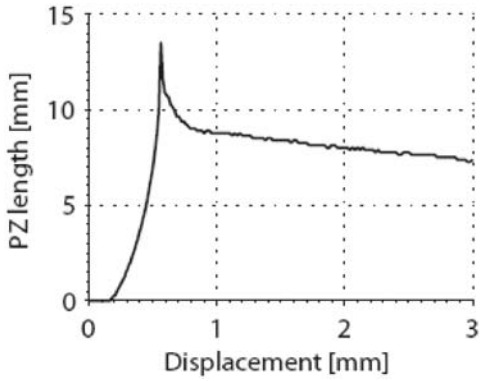


Figure 6. Process zone length, measured as, the vertical distance between the black and red lines in Figure 5(b), as a function of loading displacement (small-scale bridging length of Equation (1), $l_{SSB} = 13$ mm).

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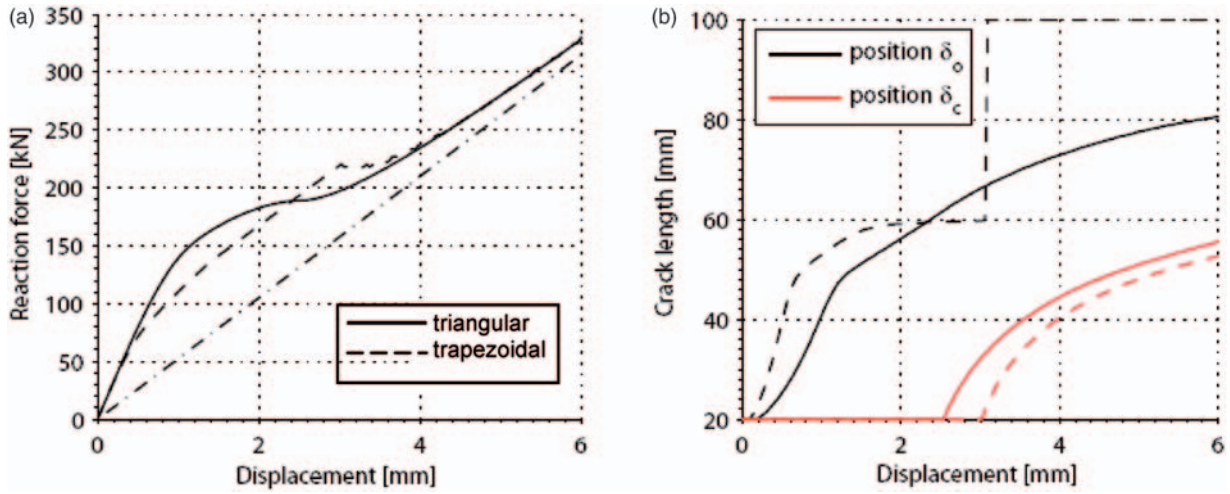


Figure 7. Results for a mode II ENF specimen loaded under quasi-static conditions with $2L = 100$ mm, $a_0 = 20$ mm, and $G_b = 5.0$ kJ/m^2. Cohesive laws correspond to fracture properties with reinforcements where the solid line represents the triangular cohesive law and the dashed is the trapezoidal.

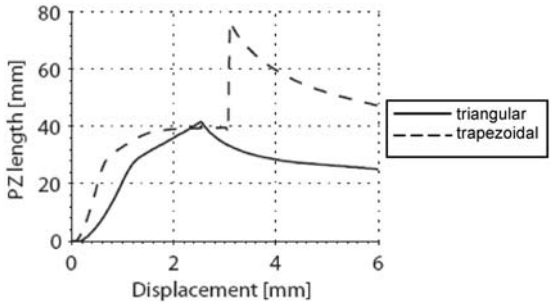


Figure 8. Process zone length corresponding to results described in Figure 7 (small-scale bridging lengths of Equation (1): $l_{SSB} = 41$ mm (triangular) and $l_{SSB} = 58$ mm (trapezoidal)).

suddenly changes slope and from here the crack tip moves at a monotonically decreasing rate relative to the load point displacement. The process zone size vs. load point displacement is plotted as the solid line in Figure 8. The history of the zone length is qualitatively similar to the nonreinforced case plotted in Figure 6, except that the zone is much longer (l_{SSB} of Equation (1) is 41 mm).

The specimen with a trapezoidal cohesive law, which is represented by dashed lines in Figures 7 and 8, has a different behavior. As the critical stress σ_0 is only half that for the triangular law, the process zone develops at a smaller load and the specimen is more compliant during the first 2 mm of loading displacement, see Figure 7(a). In addition, when the applied displacement reaches 1.2 mm, a second process zone forms at the unnotched end of the specimen. This is shown in Figure 9(a), which depicts the stresses in the normal and tangential directions along the cohesive elements in the intact ligament of the specimen for an applied displacement of 1.2 mm. In the left part of the specimen ($z < 50$ mm), the process zone has developed from the precrack for both cohesive laws, as stresses have reached the critical values $\sigma_0 = 20$ and 10 MPa, respectively. In the right half ($z > 50$ mm), the specimen is undamaged, but the specimen with the trapezoidal law (dashed line) is close to initiating a second process zone, as the shear stress σ_{zy} is approaching -10 MPa. In Figure 9(b), the corresponding opening and sliding displacements are shown.

For an applied displacement larger than 1.2 mm, the second process zone grows and propagates back toward the first crack until the two cracks join at an applied displacement of 3 mm. This is when the apparent crack tip instantly moves from 60 to 100 mm in Figure 7(b). Figure 9(c) shows the stress distribution in the cohesive elements at a loading displacement of 3 mm. The two cracks completely cover the left and right part of the specimen, where the stresses are constant and equal to $+10$ or -10 MPa (the critical traction value of the trapezoidal cohesive law). The load situation shown is the limiting case just before the two process zones join, which occurs as the sliding displacement of the precrack tip exceeds the critical displacement $\delta_c = 0.05$ mm at $z = 20$ mm, see Figure 9(b).

The history of the process zone length in Figure 8 (dashed curve) is influenced by the formation and growth of multiple process zones at applied displacements larger than 1.2 mm. The zone length shows a plateau at around 40 mm, which is well below the small-scale bridging length of Equation (1), $l_{SSB} = 58$ mm for the trapezoidal law.

The formation of multiple process zones could also be observed in the presence of triangular cohesive laws as well as in unreinforced specimens if the length of the

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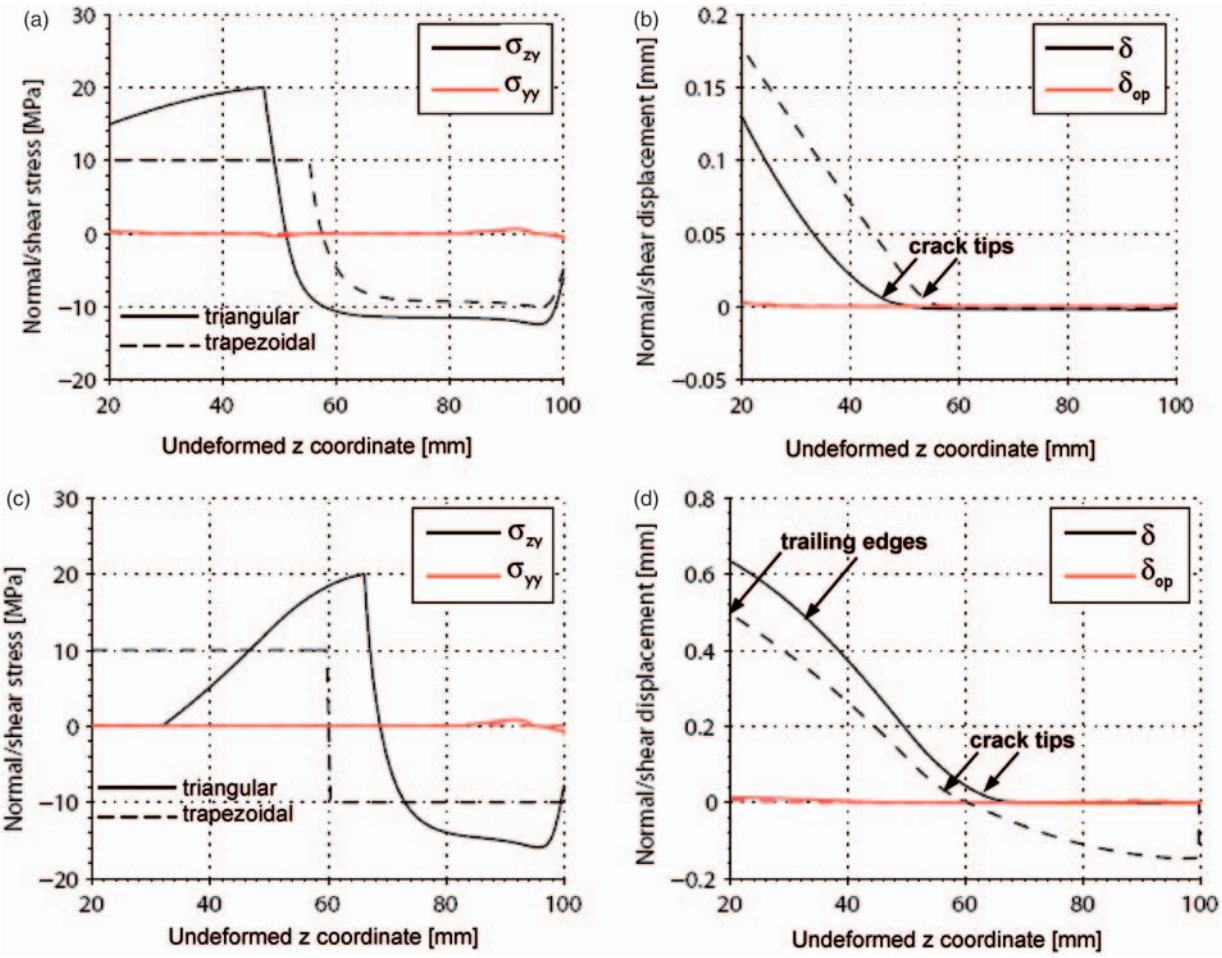


Figure 9. (a) Stresses (σ_{zy}, shear, and σ_{yy}, normal) and (b) relative displacements (δ, sliding, and δ_{op}, opening) along the cohesive elements at an applied displacement of 1.2 mm in the simulation presented in Figure 7. The prenotch tip is located at $Z = 20$ mm. The solid line represents the triangular cohesive law and dashed the trapezoidal. (c) and (d) are solutions at an applied displacement of 3 mm.

notch were much shorter, so that the notch intensification effect would become negligible.

Effect of increasing specimen length

If the specimen length L and the precrack length a_0 are increased while a fixed ratio a_0/L is maintained, the crack propagates at a smaller load and the response becomes more brittle.[10] As the nominal shear stress at the mid-plane is proportional to the force applied to the loading point, the shear stress in the intact ligament distant from the precrack may not exceed the critical stress σ_0 before the crack has propagated throughout the specimen. Results for specimens with through-thickness reinforcement and length $2L = 210$ mm are plotted in Figure 10. In this specimen, the process zone lengths rise well above the small-scale bridging limit for both cohesive laws when the crack is propagating in the first half of the specimen (approaching 60 and 72 mm for the triangular and trapezoidal laws, respectively). They then decrease while the crack propagates in the second half to 32 and 40 mm, respectively. The process zone length for the trapezoidal cohesive law is smaller than the small-scale bridging value of Equation (1), 58 mm, at the end of the simulation. This is due to nonuniformity in the stress and displacement fields, as remarked above.

In summary, from the observations of Sections 'Laminate with through-thickness reinforcement' and 'Effect of increasing specimen length', the formation of multiple process zones in the ENF specimen is favored when the cohesive law is trapezoidal or an increasing function of the relative crack displacement and when the geometry and the parameters of the cohesive law are such that the small-scale bridging length is comparable to the half-length of the specimen. In both cases, crack propagation from the notch typically occurs stably, for an increasing value of the applied load, and the stresses in the intact ligament may then exceed the critical stress σ_0. The risk of this

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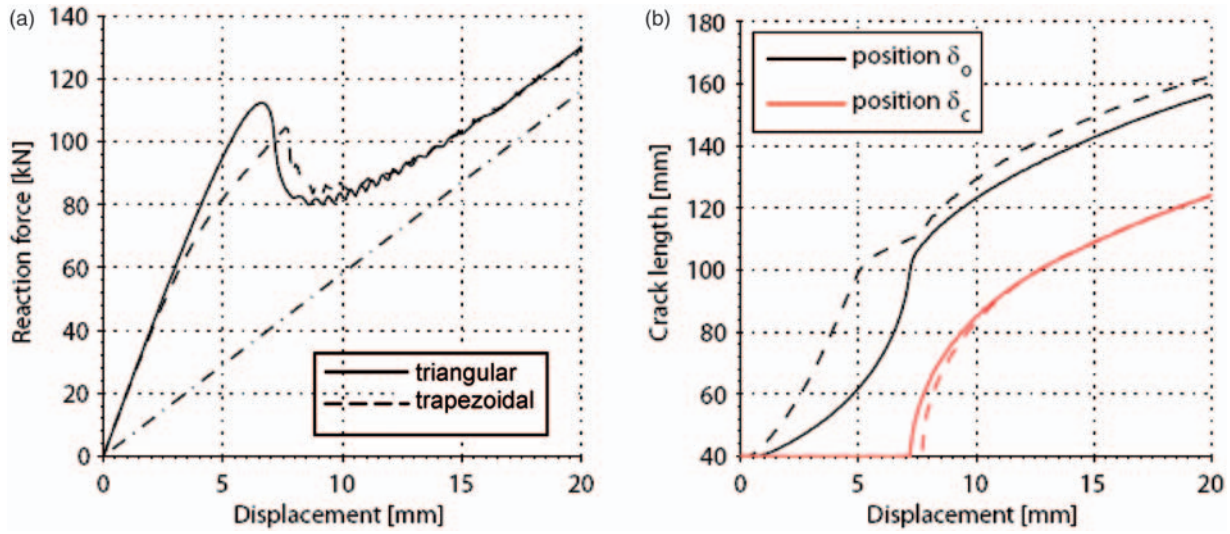


Figure 10. Quasi-static results for large specimen, $2L = 210$ mm, $a_0 = 40$ mm, and $G_b = 5.0$ kJ/m^2. Cohesive laws correspond to fracture properties with reinforcements, the solid line represents the triangular cohesive law and the dashed is the trapezoidal.

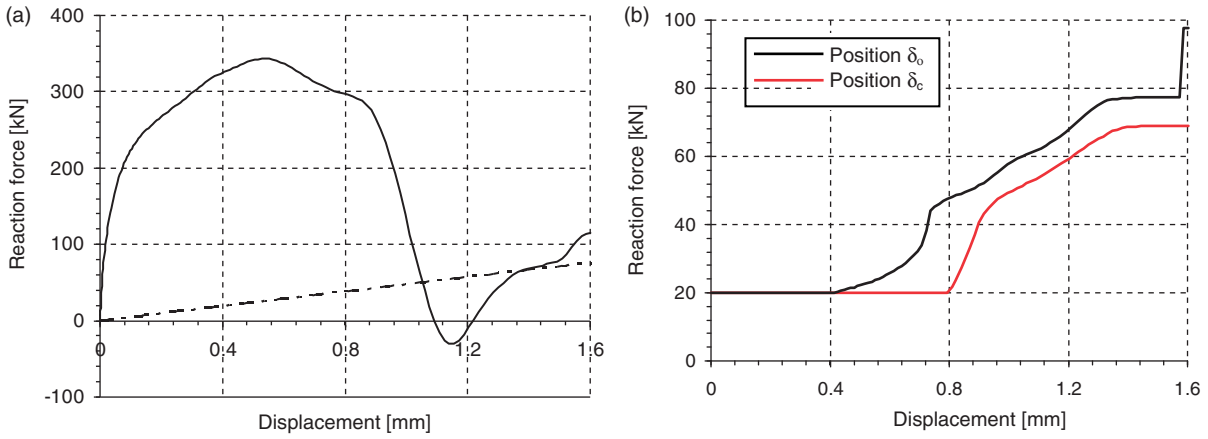


Figure 11. Results for specimen with $2L = 100$ mm, $a_0 = 20$ mm, $v = 20$ m/s, and $t_0 = T_1/2$. Cohesive law is triangular corresponding to no reinforcement ($\delta_c = 0.05$ mm). In both subplots, the horizontal axis represents the load point displacement. (a) Reaction force at loading point and (b) position of leading and trailing edges of the process zone.

phenomenon, whose presence would complicate the inference of bridging laws, may be reduced by increasing the lengths of crack and specimen, as shown in Figure 10, or reducing the thickness, and consequently increasing the ratio a_0/h, therefore, favoring a more brittle response.

Results for high-speed loading

In the following sections, results are presented for specimens with and without through-thickness reinforcement, with trapezoidal and triangular cohesive laws, loaded at high speed. The simulations follow crack growth across the entire specimen, ending when the crack breaks through its far end. This event, also

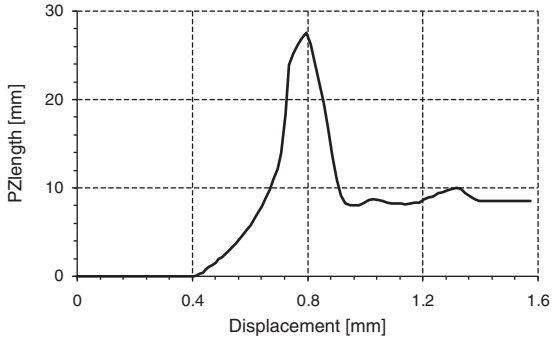


Figure 12. Process zone length, measured as, the vertical distance between the black and red lines in Figure 11, as a function of loading displacement (small-scale bridging length $l_{SSB} = 13$ mm).

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known as 'crack run-off', has been observed experimentally in ENF specimens loaded at high strain rates.[44] A simple method is suggested for relating the process zone sliding speed to the crack tip speed and displacement profile.

Laminates without through-thickness reinforcement loaded at high speed

Figures 11 and 12 refer to a specimen without through-thickness reinforcement, a load point velocity $v_0 = 20\,\text{m/s}$ and a ramp time $t_0 = T_1/2$. Comparison with the quasi-static case in Figure 5 reveals dynamic effects. For the reaction force (Figure 11(a)), dynamic effects consist of amplifications and oscillations; the process zone length (Figure 12) builds up to a much larger value (27 mm) than in the quasi-static case (13 mm) but then quickly approaches an approximately constant value of 8.5 mm (the small-scale bridging length of Equation (1) is 13 mm). In the crack propagation history (Figure 11(b)), dynamic effects induce an interval, between 1 and 1.2 mm, where the crack tip speeds remain constant and equal to 1015 m/s, which is approximately 20% of the longitudinal wave speed of the material. In this interval, the crack profile remains approximately constant and crack propagation is approximately under steady-state conditions (this will be elaborated in Section 'Sliding velocity in the process zone'). Similar intervals appear for other loading rates between 5 and 30 m/s. It will be shown below that steady-state crack propagation domains can give important information about possible rate dependence in the cohesive law. For loading rates above a certain velocity (equal to 30 m/s for this geometry, cohesive law, and loading ramp time), multiple process zones may develop along with increased oscillations, which would make experimental results more difficult to interpret.

Laminates with through-thickness reinforcement loaded at high speed

Figure 13 shows results for a specimen with through-thickness reinforcement for load point velocity $v_0 = 20\,\text{m/s}$ and ramp time $t_0 = T_1$. Comparison with the quasi-static case in Figure 7 reveals dynamic effects for the reaction force in Figure 13(a), in the form of relatively large oscillations, whereas the crack length

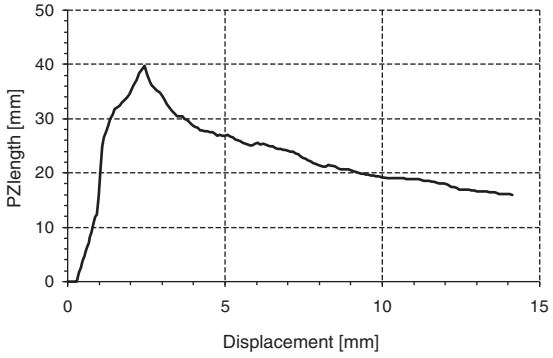


Figure 14. Process zone length, measured as, the vertical distance between the black and red lines in Figure 13, as a function of loading displacement (small-scale bridging length after Equation (1), $l_{SSB} = 41$ mm).

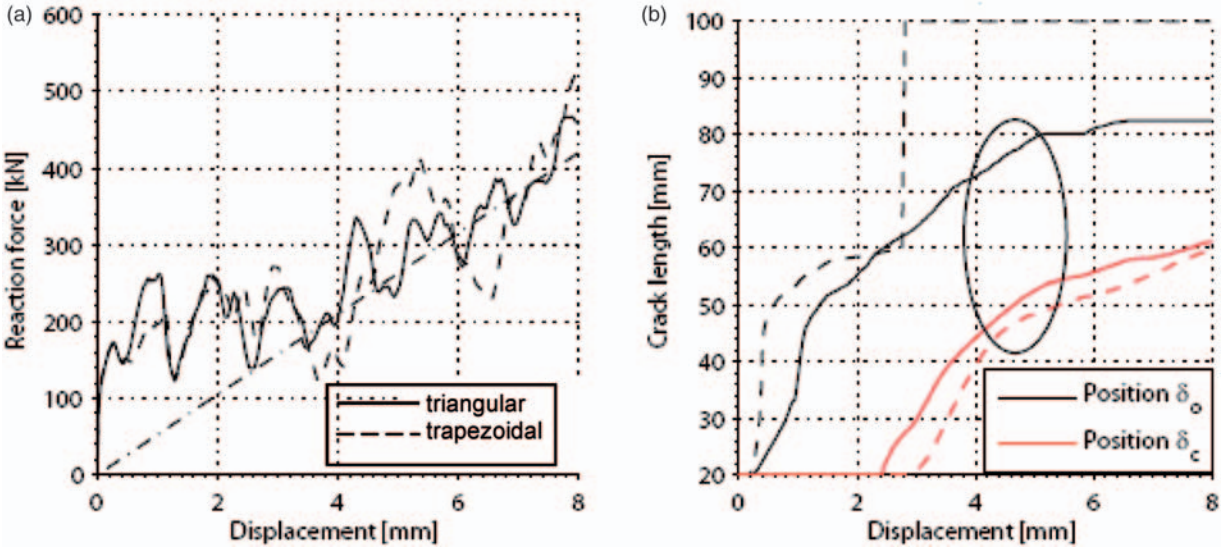


Figure 13. Results for specimen with $2L = 100$ mm, $a_0 = 20$ mm, $v = 20$ m/s, and $t_0 = T_1$. Cohesive laws correspond to a z-pinned laminate, the solid line represents the triangular cohesive law and the dashed is trapezoidal.

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history in Figure 13(b) seems less affected. In the loading displacement interval of 4.0–5.0 mm, see Figure 13(b), the leading and trailing edges move at approximately the same constant velocity of 158 m/s, and furthermore, the sliding displacement profile is also approximately constant. This will be elaborated in the following Section. The progression of the process zone length in this simulation is presented in Figure 14. After substantial variation during early propagation, when large-scale bridging conditions prevail, the zone length tends at the right of the figure toward an approximately constant value, as the small-scale bridging limit is approached. Even then, the zone length is only approximately indicated by the analytical estimate of Equation (1), $l_{SSB} = 41$ mm.

Sliding velocity in the process zone

The local sliding speed in the process zone, $\dot{\delta}(\xi)$, is a useful rate measure, as it describes the rate of the fracture processes (i.e., the rate at which through-thickness reinforcement such as z-pins or stitches are loaded, failed, and pulled out of the laminate). However, $\dot{\delta}(\xi)$ is difficult to measure experimentally. For steady-state crack propagation, namely propagation under conditions of time invariance for the crack displacement profile in the process zone and crack velocity, the local sliding speed in the process zone is related to the crack displacement profile and crack tip speed, which are experimentally measurable quantities using a digital image correlation (DIC) system, through the following relationship:

$$\dot{\delta}(\xi) = \frac{\partial \delta}{\partial t} = -\frac{\partial \delta(\xi)}{\partial \xi} v_{\text{tip}} \tag{2}$$

where ξ is a longitudinal coordinate with origin at the crack tip, v_{tip} is the crack tip velocity, and δ is the sliding displacement (Figure 15(a)).

Figure 16 shows local sliding speeds for the specimens examined in Figures 11 and 13 in the domains where steady-state conditions are suggested to hold at least approximately by the predicted crack tip speeds and crack profiles. The figure compares sliding velocities computed from simulations by (a) the right-hand side of Equation (2), where the displacement profile $\delta(\xi)$ of the process zone at time t_n is differentiated with regard to the ξ-direction and multiplied by the crack tip speed and (b) the difference in displacement between two time steps divided by the time step, that is, $[\delta(\xi, t_n) - \delta(\xi, t_{n-1})]/(t_n - t_{n-1})$, where $t_n - t_{n-1}$ is the time step (equal to 1/200 of the total time domain).

The sliding speed profiles computed by the simulation and those estimated using Equation (2) correlate well. The sliding speed varies within the process zone

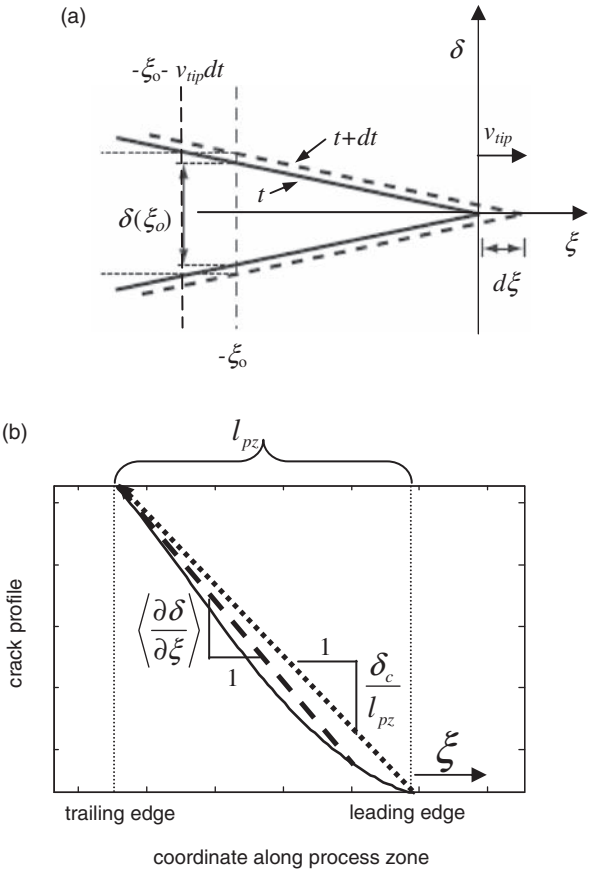


Figure 15. (a) Schematic diagram of sliding profile in steady-state conditions. (b) Exemplary sliding profile in the process zone: numerical solution (solid line), average profile along relevant portion of the process zone (dashed line), and estimate of average profile obtained joining leading and trailing edges (dotted line).

from a minimum at the leading edge to a maximum at the trailing edge. The horizontal solid lines in the figure show the average sliding speeds calculated over the process zone. The 20% of the process zone closest to the crack tip is excluded from the average, because the profile tends to be highly nonlinear there (see Figure 15(b)). The near-tip domain is also difficult to measure accurately in an experiment and, therefore, is often a weak source of information. The average sliding speed, $\dot{\delta}_{\text{ave}}$, estimated using Equation (2) when crack growth is steady state is given by

$$\dot{\delta}_{\text{ave}} = -\left\langle \frac{\partial \delta(\xi)}{\partial \xi} \right\rangle v_{\text{tip}} \tag{3a}$$

where $\langle \ldots \rangle$ denotes an average over the relevant part of the process zone. The average sliding speed obtained from the exact numerical results of simulations will equal $\dot{\delta}_{\text{ave}}$ when crack growth is steady state; in

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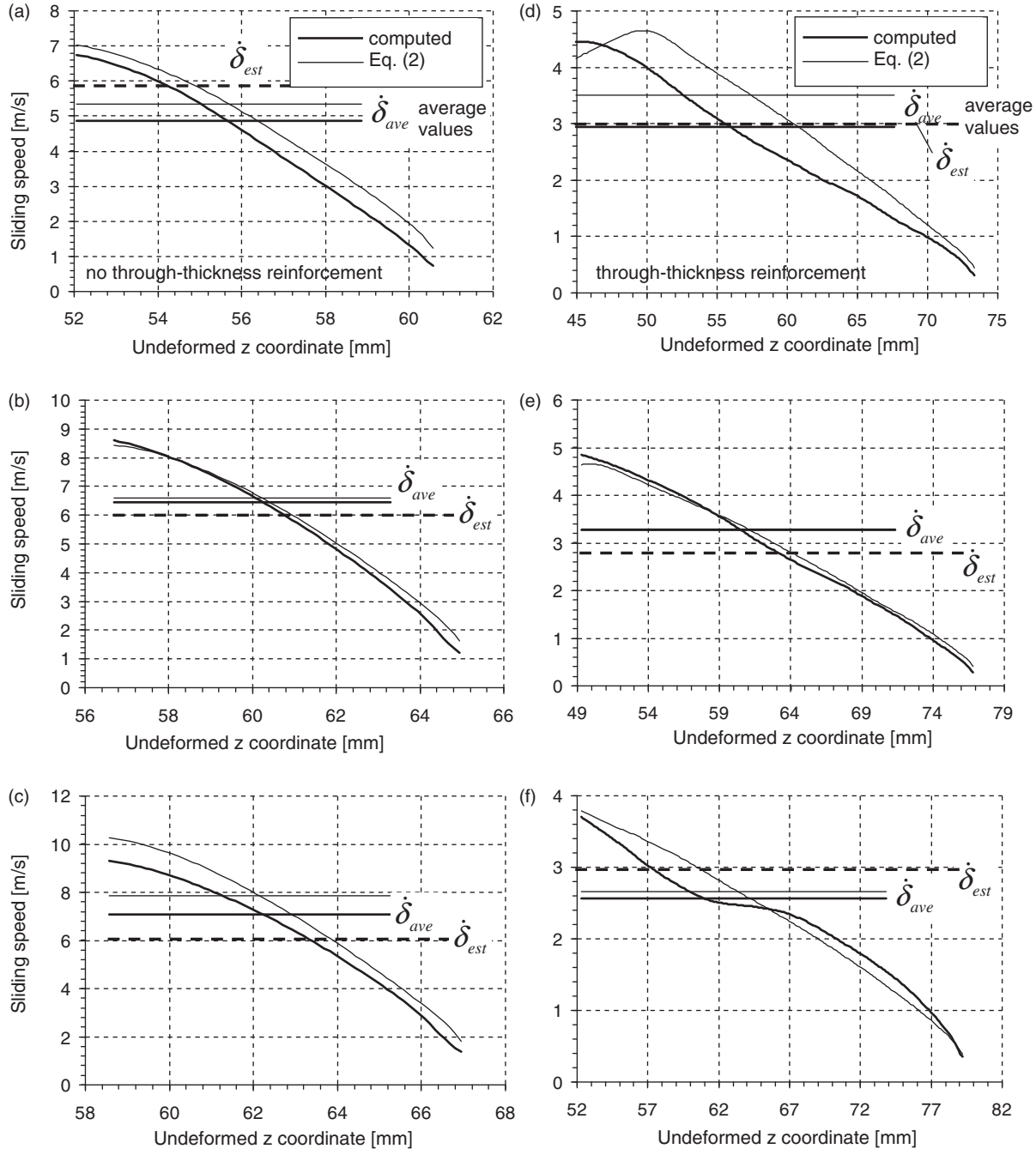


Figure 16. Actual relative sliding speeds in the process zone (from direct numerical calculations) and those predicted by Equation (2). Diagrams (a–c) correspond to results in Figure 11 at load point displacements of 1.06, 1.15, 1.19 mm; diagrams (d–f) to results in Figure 13 at displacements 4.1, 4.6 and 4.9 mm. The horizontal solid lines define average sliding speeds calculated over the relevant part of the process zone; the dashed lines define average estimates as given in Equation (3b).

addition, should the crack profile be linear, the local sliding speed along the process zone will be uniform and equal to the average $\dot{\delta}_{\text{ave}}$ everywhere.

Another useful estimate of the average sliding rate within the process zone in steady-state fracture is given by

$$\dot{\delta}_{\text{est}} = \frac{\delta_{\text{c}}}{l_{\text{pz}}} v_{\text{tip}} \tag{3b}$$

where l_{pz} is the process zone length, with $l_{\text{pz}} = l_{\text{SSB}}$ when the crack propagates in small-scale bridging conditions. For small-scale bridging and at moderate crack

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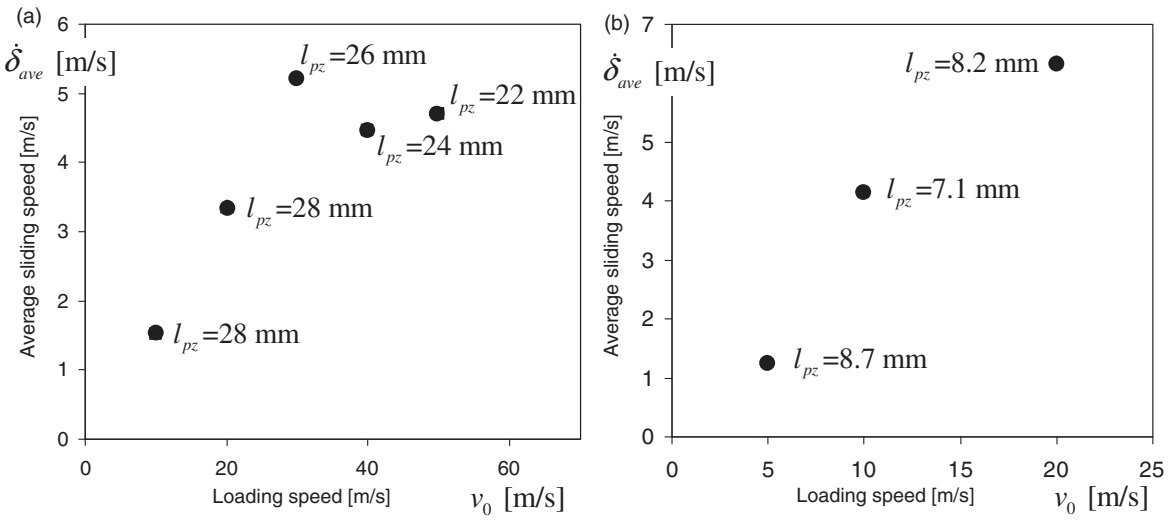


Figure 17. Average sliding speeds in the process zones in domains of steady-state crack propagation in (a) specimens with through-thickness reinforcement with $G_b = 5.0\,kJ/m^2$ (small-scale bridging length of Equation (1), $l_{SSB} = 41$ mm. (b) specimens with no through-thickness reinforcement with $G_b = 0.5\,kJ/m^2$ (small-scale bridging length of Equation (1), $l_{SSB} = 13$ mm). The corresponding lengths of the process zone are given near each data point.

tip speeds, the ratio δ_c/l_{pz} becomes independent of crack speed and is a function of the fracture energy G_b only (see Equation (1)). Figure 15(b) shows how $\dot{\delta}_{est}$ compares with $\dot{\delta}_{ave}$. The estimates of Equation (3b) are shown in Figure 16 with dashed lines. These estimates prove to be more stable in the regime of steady-state growth than those obtained with Equation (3a) (horizontal solid lines), as they depend on a global measure, l_{pz}, and not on the local details of the crack profile that are more affected by dynamic effects and numerical errors. In the next section, it will be shown that $\dot{\delta}_{ave}$ and $\dot{\delta}_{est}$, when acceptably accurate, can be used to predict trends in behavior.

Figure 17 shows how average sliding speeds and process zone lengths in the domains of steady-state propagation relate to the loading speeds. The average sliding speed across the process zone during steady-state propagation increases monotonically with the loading speed provided: (a) the length of the process zone remains similar for the different loading speeds, (b) multiple process zone formation is avoided, and (c) the loading speed (and ramp time) is not so high (small) as to modify the response of the specimen (see Section 'Characteristic length scales'). These conditions are satisfied in ENF specimens when the crack is propagating under approximate small-scale bridging conditions for all loading speeds, as in the case of the specimen with no through-thickness reinforcement (Figure 17(b)). If the crack propagates under general large-scale bridging conditions or when multiple process zones develop, the length of the process zone in the steady-state domains will vary significantly with loading speed, leading to corresponding variations in the response of the specimen. For example, the average sliding speed in the steady-state domains may decrease on increasing the loading speed (Figure 17(a), specimen with through-thickness reinforcement).

The results of the numerical simulations also show that the full solution space contains quite complex behavior. Cases can be observed where steady-state propagation never occurs even if the crack is propagating under small-scale bridging conditions, because the crack tip speed changes continuously during the fracture process; this typically occurs when the loading speeds are small and in the quasi-static case (see, for instance, Figure 5). Conversely, cases can be observed when multiple domains of steady-state propagation with different sliding speeds are observed in a single simulation. These cases typically occur when the flexural oscillations of the specimen loaded dynamically cause propagation with alternating phases of crack arrest followed by phases of crack growth.

The results above indicate that measurements of the loading speed are not enough to infer rate effects in an ENF specimen, because of the variety of different possibilities for the details of the fracture process for similar loading rates. Section 'Sensitivity to rate dependence in the cohesive law' will show instead that the crack tip speed correlates with rates in the crack process zone and therefore experiments must attend to measuring the crack tip speed and preferably the crack profile as well. (In the flying wedge specimen,[35] both the crack tip speed and the length of the process zone are directly related to the loading speed, because the load is applied directly to the fracture surfaces at locations that follow the crack motion. In such a case, steady-state crack propagation may occur both in small- and large-scale bridging and the loading speed

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can be the reference parameter. However, the flying wedge is a special case for mode I loading only.)

Sensitivity to rate dependence in the cohesive law

The question of whether a particular dynamic test will yield information about possible rate dependence in the cohesive law can be addressed by analyzing the solutions presented above along with simulations that assume hypothetical rate dependence of the cohesive law parameters. Even this theoretical exercise is complex if undertaken most generally: the results from above show that in the ENF geometry, the crack sliding speed changes continually over time and along the process zone, which complicates the inverse solution of the problem. Therefore, a simplified approach is taken of restricting the domains of the solutions considered to those where steady-state conditions hold at least approximately and assuming idealized cohesive laws.

Consider a separable linear rate-dependent cohesive law of the form

$$\sigma = \sigma_t + s_t \delta \quad (\delta \leq \delta_{tc}) \tag{4a}$$

$$\sigma = 0 \quad (\delta > \delta_{tc}) \tag{4b}$$

with δ_{tc} the critical sliding displacement beyond which the cohesive tractions vanish and s_t the slope of post-critical branch. This linear law has often been used in the literature (see [47] for work by the authors) and experiments have shown it to be a useful approximation for stitches in shear[10] or tension[9] and pins in shear or tension.[11] For negative and zero values of s_t, the law defines the triangular and rectangular laws used above; $s_t < 0$ aptly describes the intrinsic cohesive mechanisms of conventional tape laminates in the absence of through-thickness reinforcement. [When the cohesive law of Equation (4) is used with cohesive interface elements, an initial linear elastic branch needs to be included as in Figure 2. The sliding displacement of Equation (4) is then given by the difference of the total displacement across the crack and the displacement at the end of the linear branch.]

Assume that rate dependence consists of linear changes in the parameters of Equation (4):

$$\sigma_t = \sigma_0 + \alpha \frac{\dot{\delta}}{c_l} \tag{5a}$$

$$s_t = s_0 + \beta \frac{\dot{\delta}}{c_l} \tag{5b}$$

$$\delta_{tc} = \delta_c + \gamma \frac{\dot{\delta}}{c_l} \tag{5c}$$

where $\sigma_0, s_0,$ and δ_c are the parameter values for the static case (Figure 2 and Table 2), α, β, and γ are dimensionless rate parameters, $\dot{\delta} = \partial\delta/\partial t$ is the local sliding rate, and c_l is the longitudinal wave speed (Table 1).

The discussion in Section 'Results for high-speed loading' reveals that the crack velocity and the crack profile are approximately invariant for certain intervals during a delamination test. Tests of the sensitivity of an experiment to rate dependence are based on analyzing the domains where the crack speed is uniform, v_{tip}, and the profile is uniform and approximately linear, so that its slope is well represented by the average slope, $\langle \partial\delta/\partial\xi \rangle$. According to Equation (3a), the product of v_{tip} and the average slope $\langle \partial\delta/\partial\xi \rangle$ defines the average sliding speed. In this regime, the capacity of an experiment for testing rate dependence is therefore indicated by the range of values of the product $\langle \partial\delta/\partial\xi \rangle v_{tip}$ that can be attained, for example, by varying the load point velocity. The product $\langle \partial\delta/\partial\xi \rangle v_{tip}$ also indicates the capacity of an experiment for testing rate dependence when the crack profile is not linear, as in the cases examined in Figure 16, as its dependence on v_{tip} gives an average measure of variations of the local sliding rates.

Numerical simulations with rate-independent cohesive laws

A number of simulations were executed with cohesive laws of the form of Equation (4) with values of the parameters σ_0, s_0, and δ_c representative of the range expected for a conventional tape laminate. The linearly rising form of Figure 2 was used for the load point displacement with various displacement rates, v_0. In each simulation, a rate-independent cohesive law was specified. However, in the domain where the crack profile is linear and the crack velocity is constant, a rate effect would simply imply different but fixed values of the parameters s_t and δ_c. By examining this domain for different cases, behavior for a rate-dependent variation such as Equation (5) can be reconstructed; a simulation is executed, the domain of constant velocity and linear displacement profile is identified, and the quantity $\langle \partial\delta/\partial\xi \rangle v_{tip}$ is evaluated for this domain. The simulation is then taken to represent the law of Equations (3) and (4) for the value $\dot{\delta}_{ave} = -\langle \partial\delta/\partial\xi \rangle v_{tip}$.

Simulations were performed on ENF specimens with $2L = 100$ mm, $h = 5$ mm, and $a_0 = 20$ mm. The loading speeds were varied between 5 and 30 m/s with ramp times $t_0 = T_1/2$ or $t_0 = T_1$. A triangular cohesive law was used with maximum cohesive tractions and fracture energy varying in the following ranges: $\sigma_0 = 10$–

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50 MPa and $G_b = 0.25-2.5\,kJ/m^2$. The small-scale bridging length scales of Equation (1), $l_{SSB} = \sqrt{\delta_c \bar{E} h/(4\sigma_0)}$, corresponding to the different laws are recorded in Table 3 along with the process zone lengths measured in the simulations in the phases of steady-state growth.

Domains of steady-state crack growth were identified by examining the curves of crack length vs. applied displacement, for example Figure 11, (which must be linear for steady-state conditions to be present), the speeds of the leading and trailing edge of the crack (which must be equal and constant), and the predictions of the average sliding speeds across the process zone obtained using the numerical results and Equation (3) (the two predictions must be equal and constant in the domains of steady-state growth). Even in some cases of steady-state propagation, large high-frequency oscillations were observed in the crack tip speeds and the calculated average sliding speeds. These oscillations are a consequence of the dynamically applied load and the numerical discretization and their presence considerably complicates the identification of the steady-state domains. Steady-state domains were identified by the conditions that the average values of the two crack tip speeds are similar, the average values of the two calculated average sliding speeds across the process zone are similar, and the relative error between the two predictions falls below a limit value (selected as 10%). Only steady-state domains, where the process zone length l_{pz} would remain similar on varying the loading speed, have been considered. This assumption allows general conclusions to be drawn from results for different loading speeds.

Average sliding speeds vs. loading and crack tip speeds

Figure 18(a) shows the average sliding speeds in the steady-state domains obtained for different cohesive laws as a function of the loading speed. The average sliding speed generally increases with the loading speed, with the interesting variation that different domains of steady-state growth can occasionally be observed for the same value of the loading speed, for example, for $\sigma_0 = 30\,MPa$ and $G_b = 0.5\,kJ/m^2$ (they occur at different stages of crack growth). This is true also for $\sigma_0 = 50\,MPa$ and $G_b = 2.5\,kJ/m^2$, where, however, for loading rates above 30 m/s the average crack sliding speed decreases.

Figure 18(b) shows the average sliding speeds vs. the crack tip speeds. Points appear in pairs, representing the leading and the trailing edges of the crack; their closeness indicates how well the steady state is attained. The process zone lengths l_{pz} are recorded in Table 3 along with the average crack tip speeds. The dashed

Table 3. Small-scale bridging length scale, $l_{SSB} = \sqrt{\delta_c \bar{E} h/(4\sigma_0)}$ (Equation (1)), for triangular cohesive laws with the parameter values used in the numerical simulations of Figure 18; numerical process zone lengths and crack tip speeds in the approximate steady-state domains of Figure 18

	l_{SSB} (mm) (Equation (1))	l_{pz}, Process zone length simulations (mm)				
		$v_0 = 5$ m/s	$v_0 = 10$ m/s	$v_0 = 20$ m/s	$v_0 = 30$ m/s	Others (m/s)
$G_b = 0.25\,kJ/m^2$, $\sigma_0 = 20\,MPa$	9.2	4.9 ($v_{tip} = 226$)	4.85 ($v_{tip} = 352$)	5.1 ($v_{tip} = 1016$)	5.3 ($v_{tip} = 838$)	5.4 ($v_0 = 15$) ($v_{tip} = 876$)
			5.0 ($v_{tip} = 459$)			
$G_b = 0.5\,kJ/m^2$, $\sigma_0 = 20\,MPa$	13	8.7	7.1	8.2	–	–
$G_b = 0.5\,kJ/m^2$, $\sigma_0 = 30\,MPa$	8.7	4.7 ($v_{tip} = 138$)	4.15 ($v_{tip} = 414$)	4.5 ($v_{tip} = 903$)	–	–
				4.7 ($v_{tip} = 1118$)		
				4.9 ($v_{tip} = 1437$)		
$G_b = 0.5\,kJ/m^2$, $\sigma_0 = 10\,MPa$	26	–	16.5	10	–	–
$G_b = 0.75\,kJ/m^2$, $\sigma_0 = 30\,MPa$	10.6	6.3 ($v_{tip} = 140$)	5.5 ($v_{tip} = 256$)	5.7 ($v_{tip} = 703$)	5.0 ($v_{tip} = 1300$)	–
			5.8 ($v_{tip} = 345$)			
$G_b = 1.0\,kJ/m^2$, $\sigma_0 = 30\,MPa$	12.3	–	7.1 ($v_{tip} = 193$)	–	7.9 ($v_{tip} = 1015$)	6.5 ($v_0 = 40$) ($v_{tip} = 348$)
$G_b = 2.5\,kJ/m^2$, $\sigma_0 = 50\,MPa$	11.6	–	7.5 ($v_{tip} = 52$)	5.7 ($v_{tip} = 351$)	8.15 ($v_{tip} = 1015$)	–
				6.3 ($v_{tip} = 580$)		

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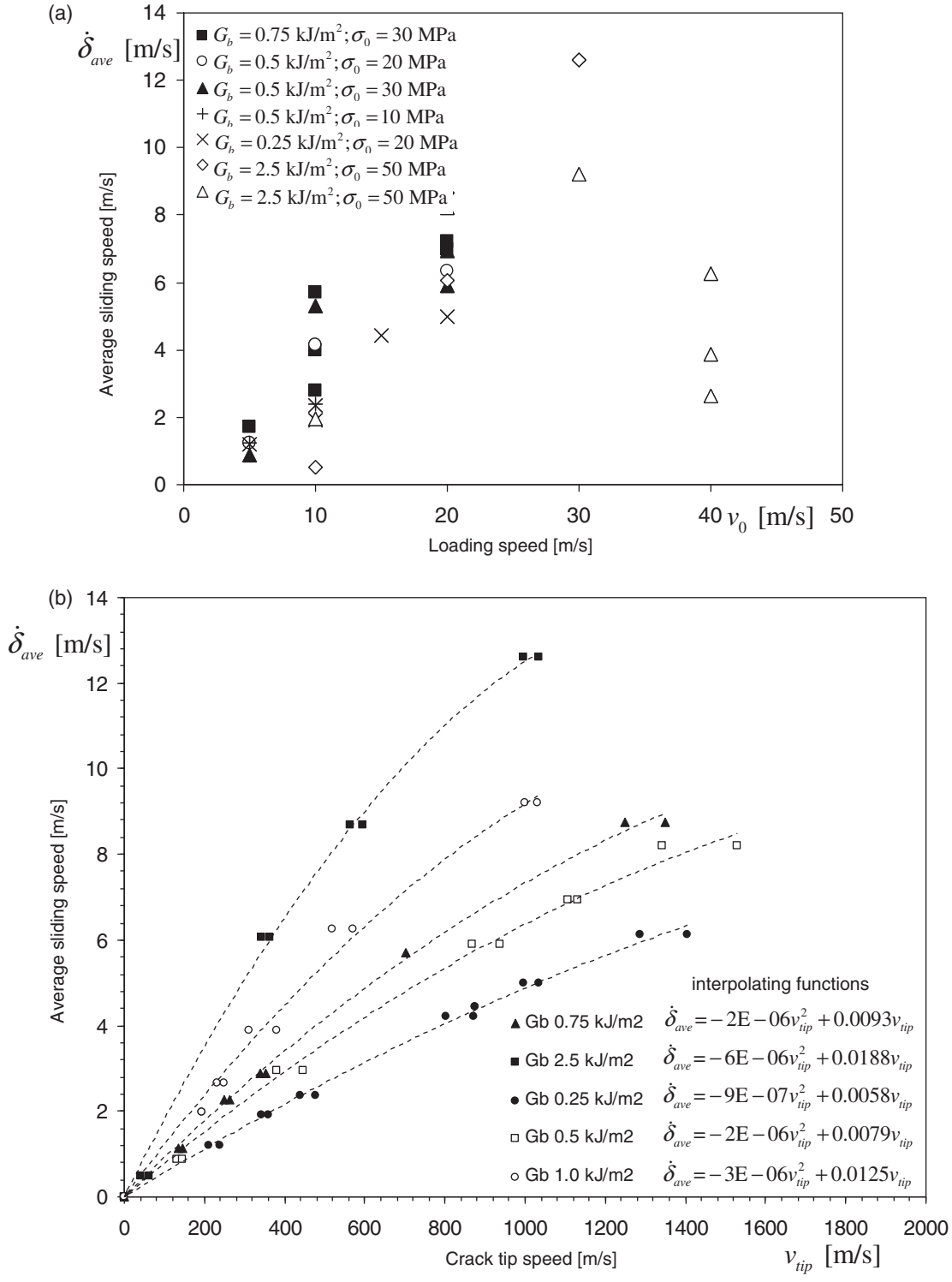


Figure 18. Average sliding speeds in the process zones in domains of steady-state crack propagation and small-scale bridging as functions of (a) the loading speed and (b) the crack tip speed for different triangular cohesive laws. The smooth dashed curves in (b) are interpolating functions used to construct plots of anticipated trends for hypothetical rate-dependent materials.

curves in Figure 18(b) are interpolating functions constrained to pass through the origin ($\dot{\delta}_{\text{ave}} = 0$, $v_{\text{tip}} = 0$). The monotonic trends of these curves is key to interpreting experiments: changing the crack tip speed causes significant and useful changes in the crack sliding speed, at least in the steady state. In addition, a linear trend at moderate speeds indicates that $\langle \partial\delta/\partial\xi \rangle$ is approximately constant on varying v_{tip},

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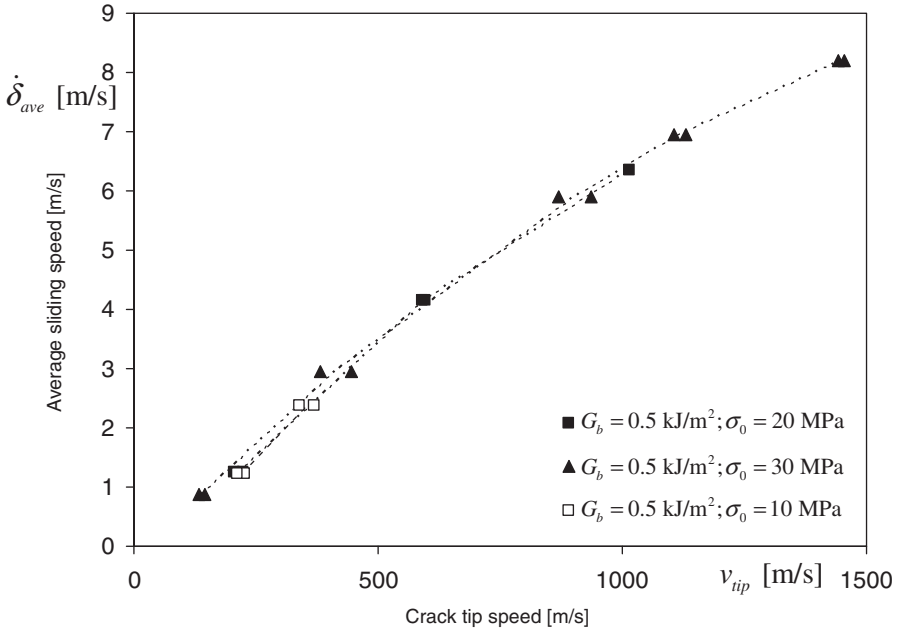


Figure 19. Average sliding speeds in the process zones in domains of steady-state crack propagation and small-scale bridging as functions of the crack tip speed for three triangular cohesive laws with fracture energy $G_b = 0.5\,\mathrm{kJ/m^2}$.

Equation (3), implying small-scale bridging conditions with an approximately constant size of the process zone length $l_{pz} = l_{SSB}$. Departure from linearity at larger speeds suggests conditions other than small-scale bridging (typically leading to larger l_{pz}) or the growing influence of kinetic energy (leading to shorter l_{pz}).

The totality of results show that a significant range of crack sliding speeds can be generated in dynamic ENF tests for loading rates typical of a split Hopkinson bar system,[43] encouraging the expectation that any rate dependence in cohesive laws can be determined.

Figure 19 shows average sliding speeds in the steady-state domains vs. the crack tip speed for three different cohesive laws possessing the same fracture energy $G_b = 0.5\,\mathrm{kJ/m^2}$. The results align closely on a single curve, which is approximately linear for crack tip speeds below $0.2c_1$. This is expected because, in all cases studied, the crack propagates in the steady state with small-scale bridging conditions, where only the area under the cohesive law, G_b, matters and not its shape. Thus, to obtain information on how rate effects modify the shape of the cohesive law, data will have to be acquired when the crack is growing in large-scale bridging.

Reconstructing behavior of rate-dependent materials

For small-scale bridging conditions, any material rate dependence that manifests itself in the cohesive fracture law should only affect the fracture energy G_b through changes in the average sliding speeds, $\dot{\delta}_{ave}$ or $\dot{\delta}_{est}$ (or the crack tip speed, v_{tip}, which is proportional to $\dot{\delta}_{ave}$ and $\dot{\delta}_{est}$ in steady-state propagation). The three rate parameters of Equation (5) do not act independently; only their effect on the fracture energy influences behavior. In the case of linear rate dependence, one may write

$$G_b = G_b^{(0)}\left(1 + \theta\frac{\dot{\delta}_{est}}{10^{-3}c_1}\right) \tag{6}$$

where $G_b^{(0)}$ is the quasi-static value of G_b, $\dot{\delta}_{est}$ is the estimated average sliding speed in the process zone, Equation (3b), and θ is a dimensionless constant, whose value is kept near unity by the presence of the factor 10^{-3} at the denominator.

Figure 20(a) replots the estimated average sliding speed, $\dot{\delta}_{est}$ of Equation (3b) from the linear regime of Figure 18b (moderate tip speeds). The dashed lines are linear interpolations corresponding to different cohesive laws, constrained to pass through the origin. Their slopes define the ratio δ_c/l_{SSB} (Equation (3b)), which gives average measures of the process zone lengths of the different materials. For instance, the curve for $G_b = 0.5\,\mathrm{kJ/m^2}$ has a slope $\delta_c/l_{SSB} = 0.0073$, so that $l_{SSB} = 4.56, 6.85,$ and 13.7 mm for $\delta_c = 0.033$ (or $\sigma_0 = 30$ Mpa), 0.05 (or $\sigma_0 = 20$ Mpa) and 0.1 mm (or $\sigma_0 = 10$ Mpa), respectively. Comparing these predictions with the predictions of Equation (1), shown in Table 3, highlights the influence of the nonuniform tractions acting along the crack: the numerical process zone length is approximately $0.53\ l_{SSB} = 0.53\sqrt{\delta_c \bar{E} h/(4\sigma_0)}$.

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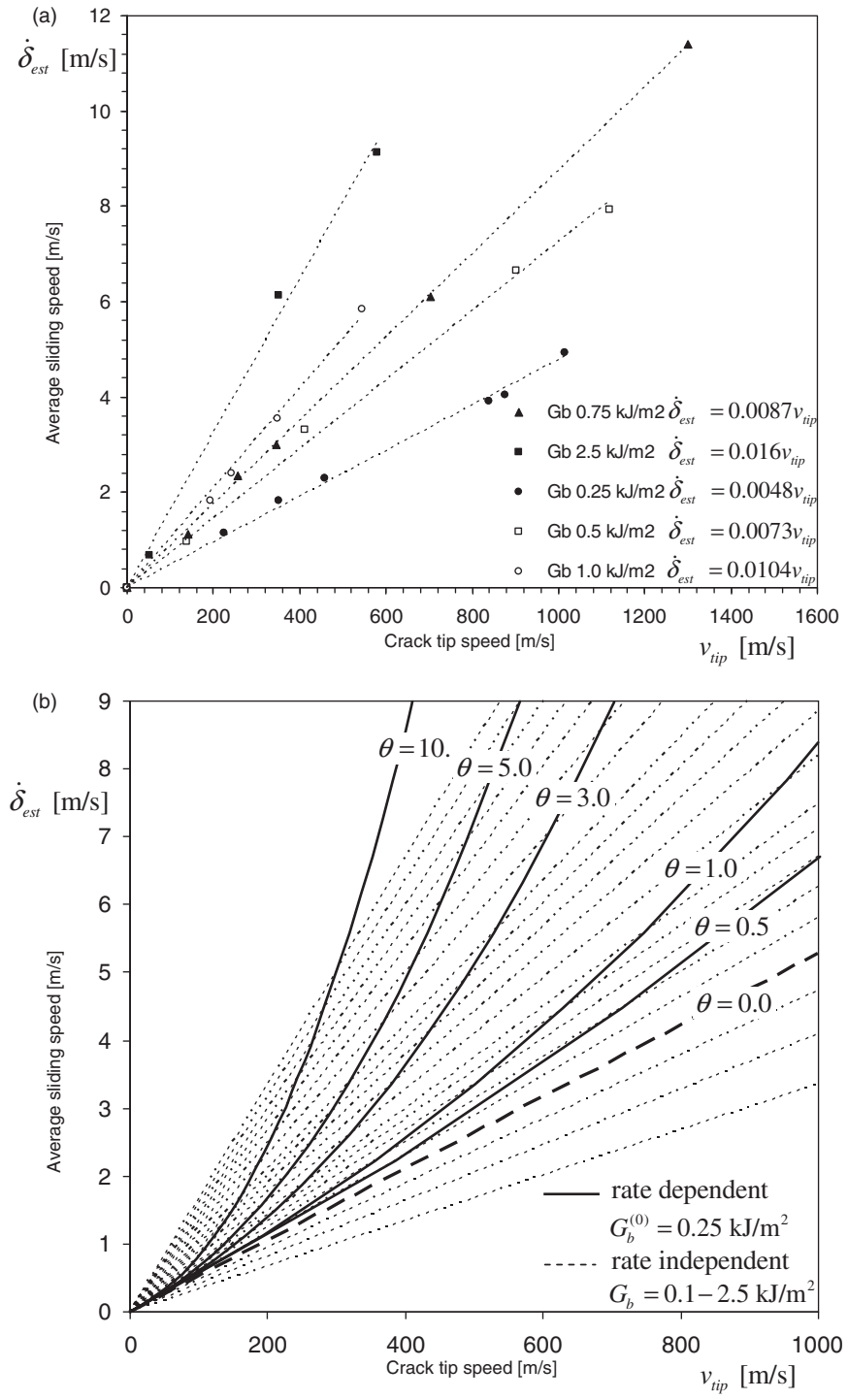


Figure 20. (a) Average sliding speeds predicted within the process zones in domains of steady-state crack propagation as functions of the crack tip speeds; the dashed lines are interpolating functions. (b) Average sliding speeds predicted within the process zones in domains of steady-state crack propagation and small-scale bridging as functions of the crack tip speeds for rate-independent materials dashed lines (Equation (3b)) and hypothetical rate-dependent materials (solid curves) with quasi-static fracture energy $G_b^{(0)} = 0.25\,\text{kJ/m}^2$ and rate dependence given by Equation (6).

The dashed lines in Figure 20(b) summarize the response of rate-independent materials propagating in small-scale bridging in the steady-state regime through the function $\dot{\delta}_{\text{est}} = v_{\text{tip}}\delta_c/l_{\text{SSB}}$, with $l_{\text{SSB}} = 0.53\sqrt{\delta_c \bar{E} h/(4\sigma_0)}$ and $G_b = G_b^{(0)} = 0.1 - 2.5\,\text{kJ/m}^2$. These curves can be used to construct steady-state behavior

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for a hypothetical rate-dependent material. For the examples shown, the rate-dependent material is assumed to have the linear rate dependence of Equation (6) and $G_b^{(0)} = 0.25\,\mathrm{kJ/m^2}$. Each solid curve in the figure describes the response for an assumed value of the rate parameter θ; the curve intersects the dashed curves corresponding to the different G_b at the average sliding speed $\dot{\delta}_{est}$ given by Equation (6). The thick dashed curve with $\theta = 0$ defines the response of a rate-independent material with $G_b^{(0)} = 0.25\,\mathrm{kJ/m^2}$.

For low crack tip speeds, the influence of rate effects on the average sliding speed is limited: when $v_{tip} < 50\,\mathrm{m/s}$, variations in $\dot{\delta}_{est}$ due to rate effects remain below 10% even if the rate parameter θ is varied substantially from 0.0 to 10 ($\theta = 10$, for the material tested in this study gives $G_b = G_b^{(0)}(1 + 1.8\dot{\delta}_{est})$). The same is true for all $v_{tip} < 100\,\mathrm{m/s}$ when θ remains below 5.0 (which corresponds to $G_b = G_b^{(0)}(1 + 0.9\dot{\delta}_{est})$). For higher crack tip speeds, rate dependence in the cohesive law can be readily observed in the numerical solutions (and therefore, in experiments). For a crack tip speed of 400 m/s, for instance, the average sliding speed of 2 m/s of a rate-independent material, increases to 2.5 m/s when $\theta = 1$ (and $G_b = 0.4\,\mathrm{kJ/m^2}$) and to 3.5 m/s for $\theta = 3$ (and $G_b = 0.7\,\mathrm{kJ/m^2}$). A rough guide for test design is inferred to be that the test should achieve crack velocities in excess of 100 m/s. However, as observed in the numerical simulations performed above, only crack speeds below certain limit values are accessible in the tests as increasing the loading speed above certain values typically leads to multiple crack formation and other complicating mechanisms that might be advantageous for engineering performance but are unwelcome in tests intended to calibrate rate dependence. The limit values depend on the geometry of the specimen and the elastic and inelastic properties of the material.

Maps similar to that shown in Figure 20(b) can be constructed for different values of the quasi-static fracture energy $G_b^{(0)}$ and can be used to interpret experimental results in domains of steady-state growth in ENF specimens.

Optimal specimen design

The design of an optimal test procedure should address the requirements listed in Section 'Introduction,' for which the following conclusions are reached.

1. Large crack speeds should be avoided: ideally $v_{tip} < 0.2 - 0.3c_l$, where c_l is the longitudinal wave speed. For higher crack speeds, the fracture process is dominated by kinetic energy and the effects of the reinforcement on the solution become negligible and, therefore, unimportant in engineering applications and not amenable to inference from data. Furthermore, increasing the loading speed above a certain limit induces other unwelcome mechanisms (which in practice will prevent the crack tip from reaching very high speeds), in particular the formation of multiple delaminations and the development of high-order high-frequency waves (the same effect is observed by decreasing the loading ramp time).

2. Very low crack speeds should be avoided as rate effects will not be easily measurable there; for instance, for crack propagation in the small-scale bridging limit and the linear rate dependence assumed in Equation (6), the crack tip speeds should be $v_{tip} > 0.02c_l$ (Figure 20(b)). The analyses show that conditions 1 and 2 should be achievable: ramp times on the order of the first period of vibration of the specimen, $t_0 = 0.5$–$1T_1$, and loading speeds in the range of those that can be applied using a split Hopkinson bar system, $v_0 = 5$–40 m/s, will yield crack tip speeds in a preferred range.

3. To generate crack propagation in the domain where the process zone is developing and also where it is fully developed, the half-length of the ENF specimen should be several times longer than the small-scale bridging characteristic length scale, which is a function of the square root of the thickness of the specimen. Furthermore, the length of the notch should be longer than a transitional value, which is inversely dependent on the small-scale bridging length scale, l_{SSB}, and directly dependent on the slenderness of the specimen, L/h.

Although these requirements can be satisfied in conventional tape laminates where, l_{SSB} remains in the range 1–5 mm, using sufficiently long specimens, in through-thickness reinforced laminates where $l_{SSB} = 40$–60 mm (in the examples shown) would lead to very long and much too compliant specimens. In general, it can be expected that, as in the static case, the information content of dynamic ENF tests may not include the regime where the through-thickness reinforcement ultimately fails. In that case, supplementary short beam shear tests might be successful in achieving reinforcement failure.

4. The formation of multiple cracks in the specimen should be avoided.

The initiation and propagation of cohesive cracks at the unnotched end of the specimen is controlled by the geometry of the specimen and the loading procedure. Multiple cracking can be avoided using longer and thinner specimens, longer notches, increasing the ramp time, and reducing the loading speed. In addition, the shape of the cohesive law has an important effect on this mechanism: multiple cracks are more likely when

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the bridging tractions are an increasing function of the relative crack displacement. The occurrence of delaminations under the applied load in the early stages of loading has not been checked in this study and would require a model with a larger number of cohesive interfaces.

5. Failure by mechanisms other than delamination fracture, namely compressive or tensile failure at the midspan, should be avoided.

To satisfy this requirement shorter and thicker specimens should be used.

6. To avoid large flexural oscillations of the specimen as well as the development of high-order, high-frequency waves in the desired range of crack speeds, ramp times, and loading speeds should be constrained as listed above, and the specimen compliance should be limited by limiting the slenderness L/h.

The dimensions used in this study, $2L = 100\,\text{mm}$ and $2h = 10\,\text{mm}$, satisfy these requirements in the assumed range of ramp times and loading speeds.

Conclusions

Through-thickness reinforcement is intended to operate under conditions where the idealizations of linear elastic fracture mechanics and small-scale bridging are incorrect; the long zones of bridging created by pins, stitches, etc., which are essential to their efficacy in delamination resistance, introduce the much richer fracture behavior of large-scale bridging conditions. In the dynamic regime, the interaction of specimen resonance, hammering fracture modes, dynamically induced multiple delaminations, and other effects leads to such a rich and complex behavioral map that the question of designing a specimen to probe the dynamic constitutive laws of the through-thickness bridging is problematic. By searching for bounds to domains of relatively simple and relatively complex behavior, this study has provided a guide to designing ENF specimens that will yield such constitutive data relatively simply.

The design was concerned with the size of the specimen (length and thickness), the length of the initial notch, and the design of the displacement-controlled loading procedure including the loading speed and the time to ramp the loading speed from zero to the required value.

The analyses show the existence of regimes, within the full computed solution space, where crack growth is approximately steady state. A simplified approach to considering rate effects in the cohesive law can be taken by restricting the domains of the solutions considered to the steady-state domains. Considering a limited solution domain is conservative with respect to assessing information content because the full domain must contain more information.

By examining the steady-state domains for different rate-independent cases, behavior for assumed rate-dependent variations can be reconstructed. In the ENF specimen, this can be done if the crack is propagating in small-scale bridging conditions; in other specimens, where the load is applied directly onto the crack surfaces, such as the mode I flying wedge specimen, behavior of rate-dependent materials can also be reconstructed for general large-scale bridging conditions using solutions for rate-independent materials.

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References

1. Cox BN, Massabò R and Kedward KT. Suppression of delaminations in curved structures by stitching. *Compos Part A* 1996; 27A: 1133–1138.
2. Glaessgen EH, Raju IS and Poe CC. Delamination and stitch failure in stitched composite joints. *AIAA J Paper 1247* 1999.
3. Holt DJ. Future composite aircraft structures may be sewn together. *Automot Eng* 1982; 90.
4. Lee C and Liu D. Tensile strength of stitching joint in woven glass fabrics. *J Eng Mater Technol* 1990; 112: 125–130.
5. Mouritz AP and Cox BN. A mechanistic approach to the properties of stitched laminates. *Compos A* 2000; 31: 1–27.
6. Rhodes MD, Williams, JG. Concepts for improving damage tolerance of composite compression panels. In: *5th DoD/NASA Conference on Fibrous Composites in Structural Design*, New Orleans, USA, January 1981.
7. Sawyer JW. Effect of stitching on the strength of bonded composite single lap joints. *AIAA J* 1985; 23: 1744–748.
8. Tada Y and Ishikawa T. Experimental evaluation of the effects of stitching on CFRP laminate specimens with various shapes and loadings. *Key Eng Mater* 1989; 37: 305–316.
9. Turrettini A. An investigation of the mode I and mode II stitch bridging laws in stitched polymer composites. Masters Thesis, Department of Mechanical and Environmental Engineering, University of California, Santa Barbara, USA, 1996.

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10. Massabò R, Mumm DR and Cox BN. Characterizing mode II delamination cracks in stitched composites. *Int J Fract* 1998; 92: 1–38.
11. Cartié DR, Cox BN and Fleck NA. Mechanisms of crack bridging by composite and metallic rods. *Compos A* 2004; 35: 1325–1336.
12. Chang P, Mouritz AP and Cox BN. Properties and failure mechanisms of *z*-pinned laminates in monotonic and cyclic tension. *Compos A* 2006; 37: 1501–1513.
13. Cox BN. Snubbing effects in the pullout of a fibrous rod from a laminate. *Mech Adv Mater Struct* 2005; 12: 85–98.
14. Dransfield K, Baillie C and Mai Y-W. Improving the delamination resistance of CFRP by stitching – a review. *Compos Sci Tech* 1994; 50: 305–317.
15. Dransfield K, Jain LK and Mai Y-W. On the effects of stitching in CFRPs-I. Mode I delamination toughness. *Compos Sci Technol* 1998; 58: 815–827.
16. Jain LK and Mai Y-W. Determination of mode II delamination toughness of stitched laminated composites. *Compos Sci Technol* 1995; 55: 241–253.
17. Jain LK, Dransfield K and Mai Y-W. On the effects of stitching in CFRPs-II. Mode II delamination toughness. *Compos Sci Technol* 1998; 58: 829–837.
18. Rugg KL, Cox BN, Ward K and Sherrick GO. Damage mechanisms for angled through-thickness rod reinforcement in carbon–epoxy laminates. *Compos A* 1998; 29A: 1603–1613.
19. Sun X, Tong L, Wood MDK and Mai Y-W. Effect of stitch distribution on mode I delamination toughness of laminated DCB specimens. *Compos Sci Technol* 2004; 64: 967–981.
20. Tong L, Cleaves K, Kruckenburg T and Stevens GP. Strength of RTM single lap joints with transverse stitching. *Key Eng Mater* 1998a; 137: 195–202.
21. Tong L, Jain LK, Leong KH, Kelly D and Herszberg I. Failure of transversely stitched RTM lap joints. *Compos Sci Technol* 1998b; 58: 221–227.
22. Chang P, Mouritz AP and Cox BN. Elevated temperature properties of pinned composite lap joints. *J Compos Mater* 2008; 42: 741–769.
23. Cox BN and Sridhar N. A traction law for inclined fibre tows bridging mixed mode cracks. *Mech Compos Mater Struct* 2002; 9: 299–331.
24. Andrews MG, Massabò R, Cavicchi A and Cox BN. Dynamic interaction effects of multiple delaminations in plates subject to cylindrical bending. *Int J Solids Struct* 2009; 46: 1815–1833.
25. Brandinelli L and Massabò R. Free vibrations of delaminated beam-type structures with crack bridging. *Compos Struct* 2003; 61: 129–142.
26. Liu H-Y, Yan W, Yu X-Y and Mai Y-W. Experimental study on effect of loading rate on mode I delamination of *z*-pin reinforced laminates. *Compos Sci Technol* 2007; 67: 1294–1301.
27. Cox BN, Sridhar N and Beyerlein I. Inertial effects in the pullout mechanism during dynamic loading of a bridged crack. *Acta Mater* 2001; 49: 3863–3877.
28. Sridhar N, Yang QD and Cox BN. Slip, stick and reverse slip characteristics during dynamic fiber pullout. *J Mech Phys Solids* 2003; 51: 1215–1241.
29. Zhu Y, Liechti KM and Ravi-Chandor K. Direct extraction of rate-dependent traction-separation laws for polyurea/steel interfaces. *Int J Solids Struct* 2009; 46: 31–51.
30. Guimard J-M, Allix O, Pechnik N and Thevenet P. Characterization and modeling of rate effects in the dynamic propagation of mode II delamination in composite laminates. *Int J Fract* 2009; 160: 55–71.
31. Tsai JL, Guo C and Sun CT. Dynamic delamination fracture toughness in unidirectional polymeric composites. *Compos Sci Technol* 2001; 61(1): 87–94.
32. Guo C and Sun CT. Dynamic mode I crack propagation in a carbon/epoxy composite. *Compos Sci Technol* 1998; 58: 1405–1410.
33. Aliyu AA, Daniel IM. Effect of strain rate on delamination fracture toughness of graphite/epoxy. In: Johnson WS (ed.), *Delamination and debonding of materials*, ASTM STP 876. Philadelphia, USA: ASTM, 1985, pp. 336–348.
34. Yaniv G, Daniel IM. Height-tapered double cantilever beam specimen for study of rate effects on fracture toughness of composites. In: Whitecomb JD (ed.), *Composite materials: testing and design*, ASTM STP 972. Philadelphia, USA: ASTM, 1990, pp. 241–258.
35. Sridhar N, Massabò R, Cox BN and Beyerlein I. Delamination dynamics in through-thickness reinforced laminates with application to DCB specimen. *Int J Fract* 2002; 118: 119–144.
36. Suo Z, Bao G and Fan B. Delamination *R*-curve phenomena due to damage. *J Mech Phys Solids* 1992; 40: 1–16.
37. Massabò R and Cox BN. Concepts for bridged mode II delamination cracks. *J Mech Phys Solids* 1999; 47: 1265–1300.
38. Brandinelli L and Massabò R. Mode II weight functions for isotropic and orthotropic double cantilever beams. *Int J Fract* 2006; 139: 1–25.
39. Smith E. The size of the fully developed softening zone associated with a crack in a strain-softening material – I. A semi-infinite crack in a remotely loaded infinite solid. *Int J Eng Sci* 1989; 27: 301–307.
40. Bao G and Suo Z. Remarks on crack-bridging concepts. *Appl Mech Rev* 1992; 24: 355–366.
41. Freund LB. *Dynamic fracture mechanics*. Cambridge, UK: Cambridge University Press, 1989.
42. Soden P, Hinton M and Kaddour A. Biaxial test results for strength and deformation of a range of E-glass and carbon fibre reinforced composite laminates: failure exercise benchmark data. *Compos Sci Technol* 2002; 62: 1489–1514.
43. Song B and Chen W. Loading and unloading split Hopkinson pressure bar pulse-shaping techniques for dynamic hysteretic loops. *Experiment Mech* 2004; 44: 622–627.
44. Nwosu SN, Hui D and Dutta PK. Dynamic mode II delamination fracture of unidirectional graphite/epoxy composites. *Compos B* 2003; 34: 303–316.
45. Williams TO and Addessio FL. A dynamic model for laminated plates with delaminations. *Int J Solids Struct* 1998; 35: 83–106.

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46. Camanho P, Davila C and De Moura M. Numerical simulation of mixed mode progressive delamination in composite materials. *J Compos Mater* 2003; 37: 1415–1438.

47. Carpinteri A, and Massabò R. Reversal in the failure scaling transition of brittle matrix fibrous composites, *Journal of Engineering Mechanics (ASCE)* 1997; 123(2): 107–114.

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